MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR

TYPES 2N2996 AND 2N2997

1. SCOPE

1.1 Scope.- This specification covers the detail requirements for germanium, PNP, low-power transistors, characterized for VHF and UHF amplifier applications and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Mechanical dimensions and outline.- TO-72 (see figure 1).

1.3 Absolute maximum ratings.-

P_T 1/	T_{stg}	V_{CB}		V_{CEO}		V_{EB}	I_C
T_A = 25°C		2N2996	2N2997	2N2996	2N2997		
mW	°C	Vdc	Vdc	Vdc	Vdc	Vdc	mAdc
75	-65 to +100	15	30	10	15	0.3	50

1/ Derate linearly at 1.0 mW/°C above T_A = 25°C.

1.4 Primary electrical characteristics.-

	h_{fe}		C_{obo}		$r_b'C_c$		NF	
	2N2996	2N2997	2N2996	2N2997	2N2996	2N2997	2N2996	2N2997
	V_{CE}=-6Vdc	V_{CE}=-12Vdc	V_{CB}=-6Vdc	V_{CB}=-12Vdc	V_{CB}=-6Vdc	V_{CB}=-12Vdc	V_{CB}=-6Vdc	V_{CB}=-12Vdc
	I_C = -4mAdc f = 100 Mc		I_E = 0 f = 1.0 Mc		I_E = -4mAdc f = 79.8 Mc		I_E = 4mAdc f = 200 Mc	
	---		pf		psec		db	
Min.	4.0		---	---	---	---	---	---
Max.	---		3.0	1.8	15	8.0	5.0	4.5

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 5961

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. - Requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. - The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500 and as follows:

$r_b\,'C_c$ . . . Collector-base time constant.

3.3 Design, construction, and physical dimensions. - Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. - Performance characteristics shall be as specified in 4.3 and 4.4.

3.5 Marking. - Marking shall be in accordance with MIL-S-19500 except that the Manufacturer's identification and Country of origin may be omitted from the body of the device at the option of the manufacturer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests. - Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.3 and 4.4. (Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (See 6.1).)

4.2 Quality conformance inspection. - Quality conformance inspection shall consist of the examinations and tests specified in 4.3 and 4.4.

4.3 Group A inspection. - Group A inspection shall consist of the examinations and tests specified in Table I.

4.4 Group B inspection. - Group B inspection shall consist of the examinations and tests specified in Table II.

4.5 Methods of examination and test. - Methods of examination and test shall be as specified in Tables I and II, and as follows:

4.5.1 Collector-base time constant. - $r_b\,'C_c$ is a figure of merit of the high frequency gain characteristics of a transistor. It may be determined by applying a radio frequency signal voltage of 1.0 Vac rms across the collector-base terminals and measuring the ac voltage drop, V_{eb}, with a high impedance rf voltmeter across the emitter-base terminals. By selecting a frequency of 79.8 megacycles for the 1.0 volt signal, $r_b\,'C_c$ in picoseconds can be calculated using the following formula:

$$r_b\,'C_c = 2 \times V_{eb}$$

where V_{eb} is in millivolts. Care should be exercised to insure that the direct collector-emitter header capacitance is balanced out during this measurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.

Table I - Group A inspection.

Examination or Test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min.	Max.	
Subgroup 1			10				
Visual and mechanical examination	2071						
Subgroup 2			3				
Collector to base cutoff current	3036 Cond. D	I_E = 0		I_{CBO}	---	-5.0	μAdc
2N2996		V_{CB} = -10Vdc					
2N2997		V_{CB} = -12Vdc					
Collector to base breakdown voltage	3001 Cond. D	I_C = -100 μAdc I_E = 0					
2N2996				BV_{CBO}	-15	---	Vdc
2N2997				BV_{CBO}	-30	---	Vdc
Collector to emitter breakdown voltage	3011 Cond. D	I_C = -4mAdc I_B = 0					
2N2996		pulsed test		BV_{CEO}	-10	---	Vdc
2N2997				BV_{CEO}	-15	---	Vdc
Collector to emitter breakdown voltage	3011 Cond. C	I_C = -100μAdc V_{BE} = 0					
2N2996				BV_{CES}	-15	---	Vdc
2N2997				BV_{CES}	-30	---	Vdc
Emitter to base breakdown voltage	3026 Cond. D	I_E = -100μAdc I_C = 0		BV_{EBO}	-0.3	---	Vdc
Forward-current transfer ratio	3076	I_C = -4mAdc					
2N2996		V_{CE} = -6Vdc		h_{FE}	25	500	---
2N2997		V_{CE} = -12Vdc		h_{FE}	40	500	---
Subgroup 3			3				
Small-signal short-circuit forward-current transfer ratio	3206	I_C = -4mAdc f = 1 kc					
2N2996		V_{CE} = -6Vdc		h_{fe}	35	---	---
2N2997		V_{CE} = -12Vdc		h_{fe}	50	---	---
Small-signal short-circuit forward-current transfer ratio	3306	I_C = -4mAdc f = 100 Mc		h_{fe}	4.0	---	---
2N2996		V_{CE} = -6Vdc					
2N2997		V_{CE} = -12Vdc					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.

Table I - Group A inspection—(Continued)

Examination or Test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min.	Max.	
Subgroup 3 cont'd							
Open-circuit output capacitance	3236	I_E = 0 f = 1.0 Mc					
2N2996		V_{CB} = -6Vdc		C_{obo}	---	3.0	pf
2N2997		V_{CB} = -12Vdc		C_{obo}	---	1.8	pf
Collector-base time constant (See 4.5.1)	----	I_E = 4mAdc f = 79.8 Mc					
2N2996		V_{CB} = -6Vdc		$r_b'C_c$	---	15	psec
2N2997		V_{CB} = -12Vdc		$r_b'C_c$	---	8	psec
Noise figure (See 4.5.3)	----	I_E = 4mAdc BW = 1 Mc f = 200 Mc					
2N2996		V_{CB} = -6Vdc		NF	---	5.0	db
2N2997		V_{CB} = -12Vdc		NF	---	4.5	db
		Noise source conductance = 13.3 mmho	15				
Subgroup 4							
Collector to base cutoff current	3001 Cond. D	T_A = 70°C I_E = 0		I_{CBO}	---	-100	μAdc
2N2996		V_{CB} = -10Vdc					
2N2997		V_{CB} = -12Vdc					
Forward-current transfer ratio	3076	T_A = -55°C I_C = -4mAdc					
2N2996		V_{CE} = -6Vdc		h_{FE}	12.5	---	---
2N2997		V_{CE} = -12Vdc		h_{FE}	20	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.

Table II - Group B inspection.

Examination or Test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min.	Max.	
Subgroup 1			20				
Physical dimensions	2066	See figure 1					
Subgroup 2			15				
Solderability	2026	Omit aging					
Thermal shock (temperature cycling)	1051 Cond. A						
Thermal Shock (glass strain)	1056 Cond. A						
Moisture resistance	1021						
Subgroup 3			10				
Shock (See 4.5.4)	2016	Nonoperating 1500G, 0.5 msec 5 blows each orientation: X_1, Y_1, Y_2, Z_1					
Vibration fatigue	2046	Nonoperating					
Vibration, variable frequency	2056						
Constant acceleration (centrifuge)	2006	20,000G each orientation: X_1, Y_1, Y_2, Z_1					
Subgroup 4			20				
Terminal strength (lead fatigue)	2036 Cond. E						
Subgroup 5			10				
Salt atmosphere (corrosion)	1041						
Subgroup 6			λ=5				
High-temperature life (nonoperating)	1031	T_A = 100°C					
Subgroup 7			λ=5				
Steady-state operation life	1026	T_A = 25°C P_T = 75 mW					
2N2996		V_{CB} = -7.5 Vdc					
2N2997		V_{CB} = -10Vdc					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.

Table II - Group B inspection--(Continued)

Examination or Test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min.	Max.	
Subgroup 7 con't							
End points for subgroups 2 and 3:							
Collector to base cutoff current	3036 Cond. D	I_E = 0		I_{CBO}	---	-5.0	μAdc
2N2996		V_{CB} = -10Vdc					
2N2997		V_{CB} = -12Vdc					
Forward-current transfer ratio	3076	I_C = -4mAdc					
2N2996		V_{CB} = -6Vdc		h_{FE}	25	500	---
2N2997		V_{CB} = -12Vdc		h_{FE}	40	500	---
End points for subgroups 6 and 7:							
Collector to base cutoff current	3036 Cond. D	I_E = 0		I_{CBO}	---	-10.0	μAdc
2N2996		V_{CB} = -10Vdc					
2N2997		V_{CB} = -12Vdc					
Forward current transfer ratio	3076	I_C = -4mAdc					
2N2996		V_{CB} = -6Vdc		h_{FE}	20	800	---
2N2997		V_{CB} = -12Vdc		h_{FE}	32	800	---
Small-signal short-circuit forward current transfer ratio	3306	I_C = -4mAdc f = 100 Mc		h_{fe}	4.0	---	---
2N2996		V_{CE} = -6Vdc					
2N2997		V_{CE} = -12Vdc					

4.5.2 Condition applying to fourth lead during testing.- All electrical tests shall be conducted with the fourth lead (case) connected to a common r.f. chassis ground.

4.5.3 Noise figure.- Noise figure shall be measured using the following commercially available test equipment and its associated standard test procedures or equivalent test equipment and procedures.

a. Hewlett-Packard VHF Noise Source, type 343A.
b. Hewlett-Packard Noise Figure Meter, type 342A.

A block diagram of the test equipment setup is shown on Figure 2. The test fixture shall consist of a 200 Mc tuned amplifier and biasing circuits. It should be constructed utilizing good VHF design techniques. The effective source susceptance should be tuned for each device being tested to obtain minimum noise figure. Because the HP 343A has a 50-ohm output resistance (on or off), an impedance transformer shall be used to obtain an effective source conductance of 13.3 mmho at the transistor with minimum losses. To make noise figure measurements, the noise figure meter, the noise source, transistor, and test fixture are interconnected. The noise figure meter squarewave modulates the noise source at approximately 500 cps and measures noise figure by comparing the noise output of the device under test when the noise source is off to the noise output when the noise source is on.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.

4.5.4 Shock.- The shock testing apparatus shall be capable of providing shock pulses of the peak acceleration, waveform, and pulse duration specified herein to the body of the device. The acceleration pulse, as determined from the unfiltered output of a transducer with a natural frequency greater than 10,000 cycles per second, shall be a half sine waveform with an allowable distortion not greater than ±20 percent of the peak acceleration specified herein. The pulse duration shall be measured between the points at 10 percent of the peak acceleration during rise and at 10 percent of the peak acceleration during decay. Absolute tolerance of the pulse duration shall be ±40 percent of the duration specified herein.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500.

6. NOTES

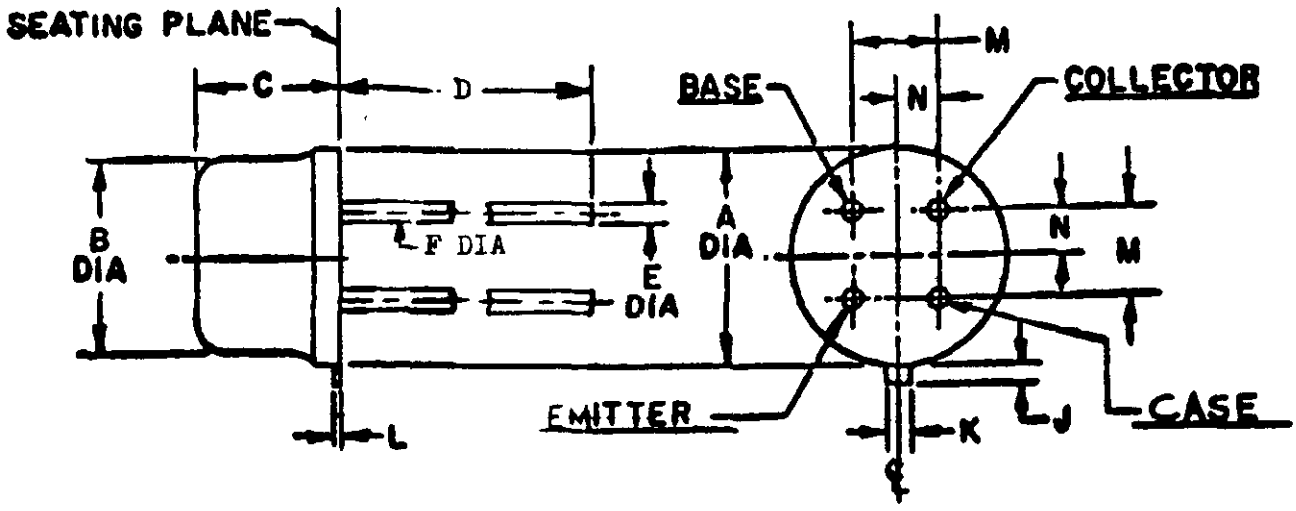
6.1 The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D.C., 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification Tests shall be made in accordance with "Provisions Governing Qualification." (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

Review activities:
Navy - SH, EC

User activites:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - SH
Project 5961-N037(NAVY))

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.



LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.209	.230	5.31	5.84	
B	.178	.195	4.52	4.95	
C	.170	.210	4.32	5.33	
D	.500	- - -	12.70	- - -	7
E	- - -	.021	- - -	.53	2,7
F	.016	.019	.41	.48	3,7
J	.028	.048	.71	1.22	6
K	.036	.046	.91	1.17	
L	- - -	.020	- - -	.51	
M	.0707 Nom		1.80 Nom		4
N	.0354 Nom		0.90 Nom		4

NOTES:
1. Metric equivalents in table and parentheses are shown for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250(6.35) from the seating plane.
3. Measured in the zone .050(1.27) and .250(6.35) from the seating plane.
4. When measured in a gaging plane $.054 \pm \frac{.001}{.000}$ $(1.37 \pm \frac{.03}{.00})$ below the seating plane of the transistor, max dia leads shall be within .007(.18) of their true location relative to a maximum width tab. Smaller dia leads shall fall within the outline of the max dia lead tolerance.
5. The emitter, base, and collector shall be electrically insulated from the case.
6. Measured from the maximum diameter of the actual device.
7. All 4 leads.

Figure 1 - Physical dimensions of transistor types 2N2996 and 2N2997.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.

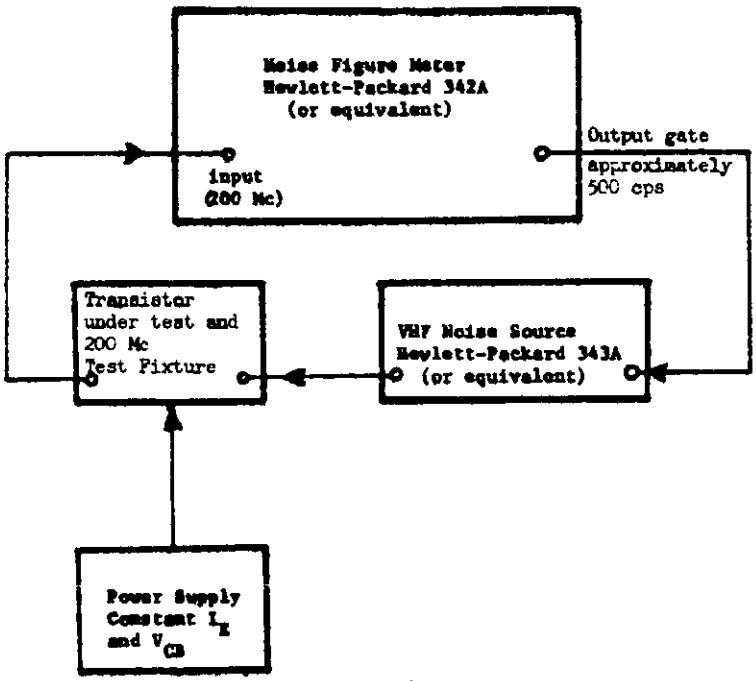


Figure 2 - Block diagram for Noise figure test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:53Z
Check the source to verify that this is the current version before use.